Proton Decay in SU(7) Grand Gauge-Higgs Unification: An Obstruction, Its Minimal Escapes, and the One Row of Their Table 1 That Can Pay for Them (Part VI)

High-energy physics · grand gauge-Higgs unification

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Statements

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Proposition 2 — a real representation under diagonal ±1 parities furnishes no chiral generation 3
Proposition 3 — the rule in closed form 4

At a glance

What the paper contributes

- An **obstruction** in the one-generation SU(7) model of Komori and Maru as published: three anomaly channels force $X_Q = -1/6$, where $U(1)' = T_{3L} + Y - (B - L)$ identically and the four dimension-6 $|\Delta B| = 1$ operators are neutral; two further channels force a right-handed neutrino of charge -1, which only a **28** supplies (rows (2) and (3) of their Table 1).
- A **classification of the minimal escapes** inside their own group: the ladder $\mathrm{extra}(L) = (1 - k)/2$, the identity $[SU(2)_L]^2 U(1)' = \frac{1}{2}\sum_j A_j$ (Proposition 1), fourteen anomaly-free three-generation assignments of which two fit their tensors; the **48** cannot host a generation because a real representation under diagonal ± 1 parities is vector-like (Proposition 2, a specialisation of Gogoladze–Mimura–Nandi).
- A **benchmark selected, not a model delivered**: $V''(0) = -\pi^2 \zeta(3) D$ with D rational per multiplet ($\mathbf{84}^{(+,+)} \mapsto 5/4$, $\mathbf{48}^{(+,+)} \mapsto 0$), so only case (2) satisfies both necessary conditions, at one-loop curvature.
- A selection rule in closed form: protection fails iff $q_\varphi = |A_j|/n$ (Proposition 3); $q_\varphi = 3/2$ protects for every half-integer brane-quark charge.
- An audit of the instrument: their $\alpha_{\min}$ column is not reproduced (stated as an open question, not a defect); the verdict holds on a wedge of multiplicative repairs; two-loop safety is left open.

🚀 **Start here: Proposition 1**: the SU(2)_L anomaly of U(1)′ is half the sum of the proton-operator charges.

Main results:

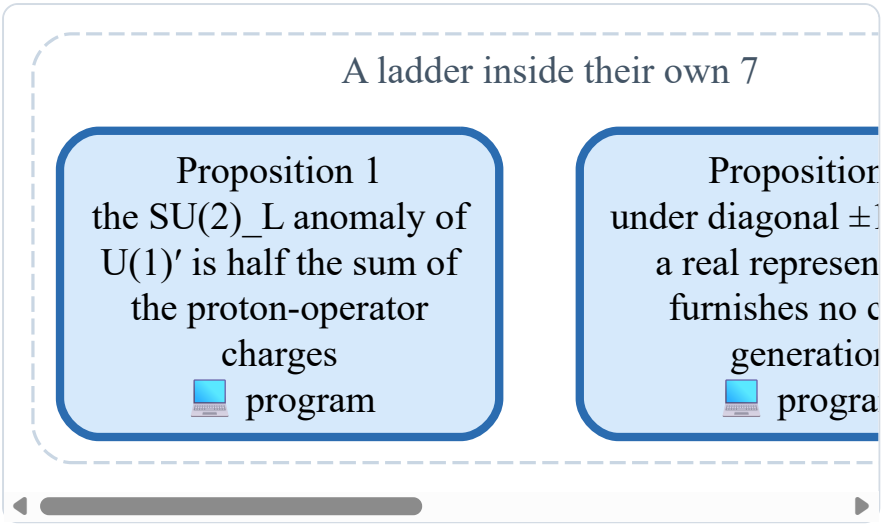
- **Proposition 1**: the SU(2)_L anomaly of U(1)′ is half the sum of the proton-operator charges
- **Proposition 2**: under diagonal ±1 parities a real representation furnishes no chiral generation
- **Proposition 3**: protection fails iff q_φ = |A_j|/n: a set with a maximum, so protection is a half-line

3 results | ✅ **0** formalized in Lean | ◐ **0** with a part in Lean

💻 **3** checked by a program | written proof only | (a result can carry several tags)

🗺️ Map of the results (click a box to go to it)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's “Evidence & scope” block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.



1 A ladder inside their own $SU(7)$

Proposition 1. (the identity that decides it) p. 9

> *For N generations with proton-operator charges $A_j = 3a_j + l_j$,*
>
> $$[SU(2)_L]^2\, U(1)' = \tfrac{1}{2}\sum_j A_j \qquad \text{exactly.}$$

> NEW **Classical and new:** the paper lists it among what is ours (§9); at N = 1 it is the entry 3X_Q/2 + 1/4 = A/2 of the anomaly table; the family-dependent cure it mirrors is Lee–Ma [FamilyU1].

↳ Not used by another result of the paper.

Evidence & scope

- **Program:** the ladder extra(L) = (1−k)/2, the identity summed state by state over N generations, the 14 anomaly-free assignments. `scripts/su7_family_u1.py`
- **Program:** the 14 assignments re-derived by an independent path, and the two that fit their tensors. `scripts/su7_realisable.py`
- **Program:** the six anomaly channels on the published one-generation content (and a Sage re-derivation, per §3). `scripts/su7_anomaly_channels.py`

Proposition 2. (a real representation under diagonal ± 1 parities furnishes no chiral generation) p. 10

> *Let the orbifold parities P_5, P_5' act diagonally on the indices of the fundamental, so that a component of a bulk multiplet survives in the multiplet of parities (η, η') iff $P_5 P_5' = \eta\eta'$ and then carries 4D chirality $-\eta\, P_5$ (their eqs. (37)–(40)). Then for a real representation the surviving spectrum is vector-like, with no chiral state at either choice of (η, η').*

> NEW **Classical and new:** not claimed as new: that adjoint matter is vector-like is stated as well known by Gogoladze–Mimura–Nandi [GMN], who cure it with a Z_3 projection; the paper claims the specialisation to the ±1 diagonal parities, the counts and the chiral controls (§9).

↳ Not used by another result of the paper.

Proof

For the adjoint $\mathbf{7} \otimes \bar{\mathbf{7}} - \mathbf{1}$ the conjugate of the component $(i, \bar{\jmath})$ is $(j, \bar{\imath})$, which lies in the *same* multiplet. Since the parities are diagonal, $P_5 P_5'(j, \bar{\imath}) = p_j p_i = P_5 P_5'(i, \bar{\jmath})$ and $P_5(j, \bar{\imath}) = P_{5,j} P_{5,i} = P_5(i, \bar{\jmath})$: the two survive together and with the same chirality. In an all-left-handed convention the two surviving Weyl fields transform in conjugate gauge representations and therefore admit a gauge-invariant Dirac mass — and their own prescription, a conjugate fixed-point fermion against each unwanted zero mode, supplies one. The argument uses no property of the Wilson line and none of the particular *values* of these parities — but it does use that they are ± 1 and diagonal, which is what makes the projection commute with conjugation, and that hypothesis is not free. A $\mathbb{Z}_3$ projection has complex eigenvalues, breaks the pairing, and is precisely how the literature obtains chiral families from an adjoint [13] . Their eqs. (37)–(40) are $\mathbb{Z}_2$, so that escape is closed in their model and not in general. □

Evidence & scope

- **Program:** the 48 at both parities: 18 of 18 and 24 of 24 surviving states paired with their conjugates; control: the 28 and the 84 leave 10 and 34 unpaired. `scripts/su7_adjoint_is_vectorlike.py`
- **Program:** the adversarial run that made the proposition necessary: the brane escape priced, every other gate open. `scripts/su7_brane_partner.py`

2 The selection rule is a half-line, not a tuning

Proposition 3. (the rule in closed form) p. 14

Protection fails at q_φ iff $q_\varphi = |A_j|/n$ for some generation j and some integer $n \geq 1$. That set is countable and has a maximum, so protection is a half-line.

NEW **Classical and new:** the mechanism (a residual discrete gauge symmetry from a broken U(1)) is Krauss–Wilczek and Ibáñez–Ross, and the leftover-U(1) breaking in gauge-Higgs unification is Dong [GHK]; the paper claims the closed-form rule for this model (§9).

↳ Not used by another result of the paper.

Evidence & scope

- **Program:** closed form against a brute scan of every rational p/q ≤ 2 with q ≤ 36, for the 14 assignments: 18 648 values, 0 disagreements. `scripts/su7_qphi.py`
- **Program:** adversarial audit: q_φ = 1/2 fails at every m, q_φ = 1 at every odd m, q_φ = 3/2 never. `scripts/su7_qphi_socratic.py`
- **Program:** the residual group and what a second condensing scalar does to it. `scripts/su7_residual_group.py`

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